Lab 3: Architectural Style Selection & Justification

Faculty Research Grant & Publication Tracker

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We chose Layered Architecture for the Faculty Research Grant & Publication Tracker System.

Architectural Choice

The system is organized into three horizontal layers: a Presentation Layer (Faculty Portal Component), a Business Layer (Grant Manager, Publication Manager, Report Generator, and Notification Service Components), and a Data Layer (Database Component). Each layer communicates only with the layer directly below it through clearly defined interfaces.

Two Reasons for This Choice

1. **Data consistency across related records:** Grant proposals, publications, and faculty profiles are tightly interrelated records that must stay consistent (e.g., a report drawing on both grant funding and publication counts for the same faculty member). A single, centrally managed Data Layer avoids the synchronization complexity that a distributed, Microservices-style data store would introduce for this scale of system.

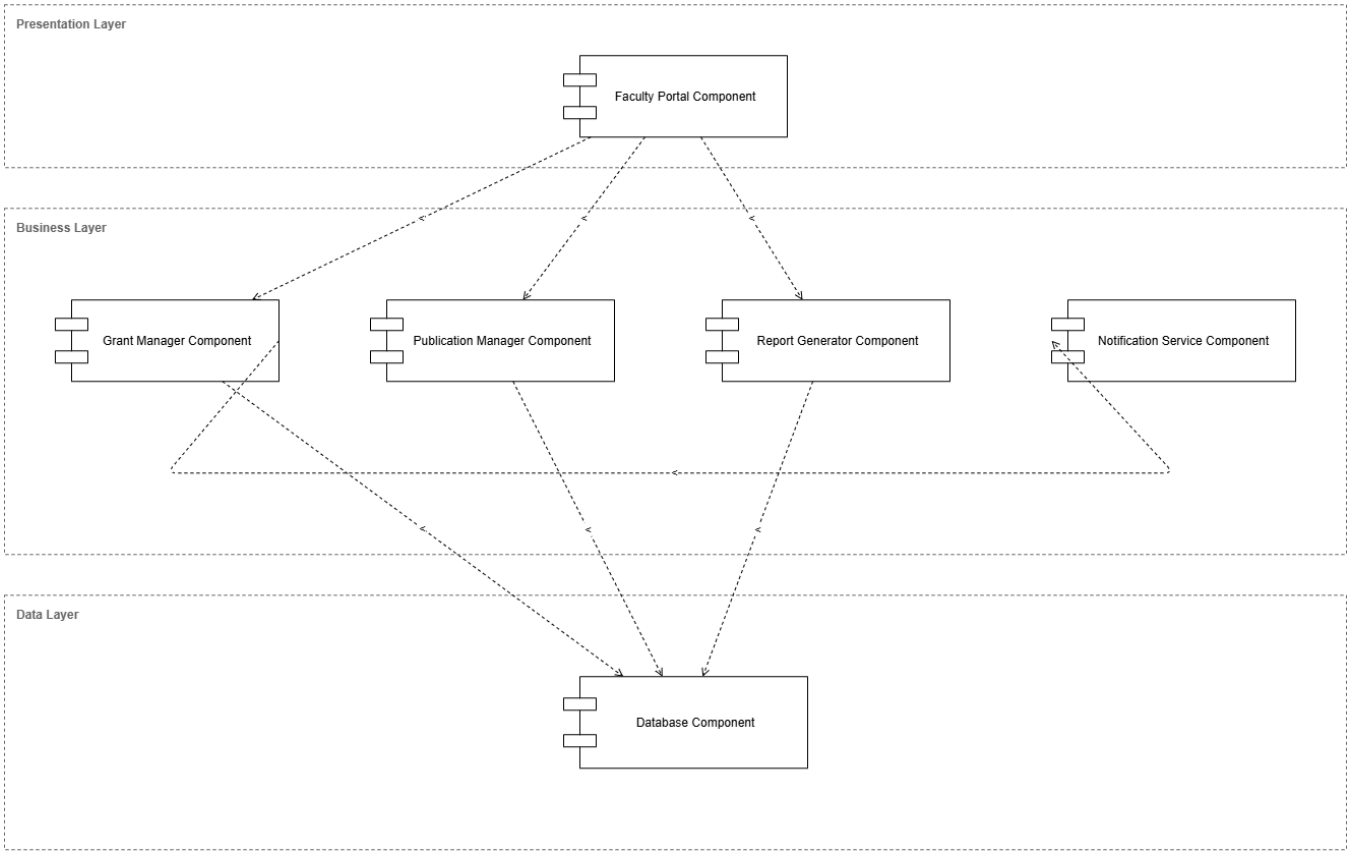
2. **Matches the scale and usage pattern of a single-institution admin system:** The tracker serves one university's faculty and research office rather than a large, globally distributed user base. Layered architecture gives clear separation of concerns (UI, business rules, data) without the operational overhead of independently deployed Microservices, and without the scalability and fault-tolerance limitations that a simple Client-Server model would impose as more report and notification features are added.

Security Advantage

Because all data access is funneled through the Business Layer (Grant Manager and Publication Manager Components) before reaching the Database Component, access control and validation logic is enforced in one place rather than scattered across the UI or duplicated per service. The Presentation Layer never talks to the Database Component directly, so sensitive grant funding data cannot be queried or modified by bypassing

business rules such as role-based permissions for faculty versus research-office administrators.

Performance Benefit

The Report Generator Component can query the Grant Manager, Publication Manager, and Database Components directly within the same layered stack, avoiding the network hops and serialization overhead that a Microservices architecture would add for every cross-component report request. For an institution-scale system with predictable, moderate load, this keeps report generation and portal response times low without needing a distributed caching or messaging layer.

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